Should I Work Out Today?

A transparent exercise decision-support experiment

AI-ASSISTED EXERCISE DECISION SUPPORT

Help people adapt, postpone or stop a planned workout when their current state no longer matches the original plan.

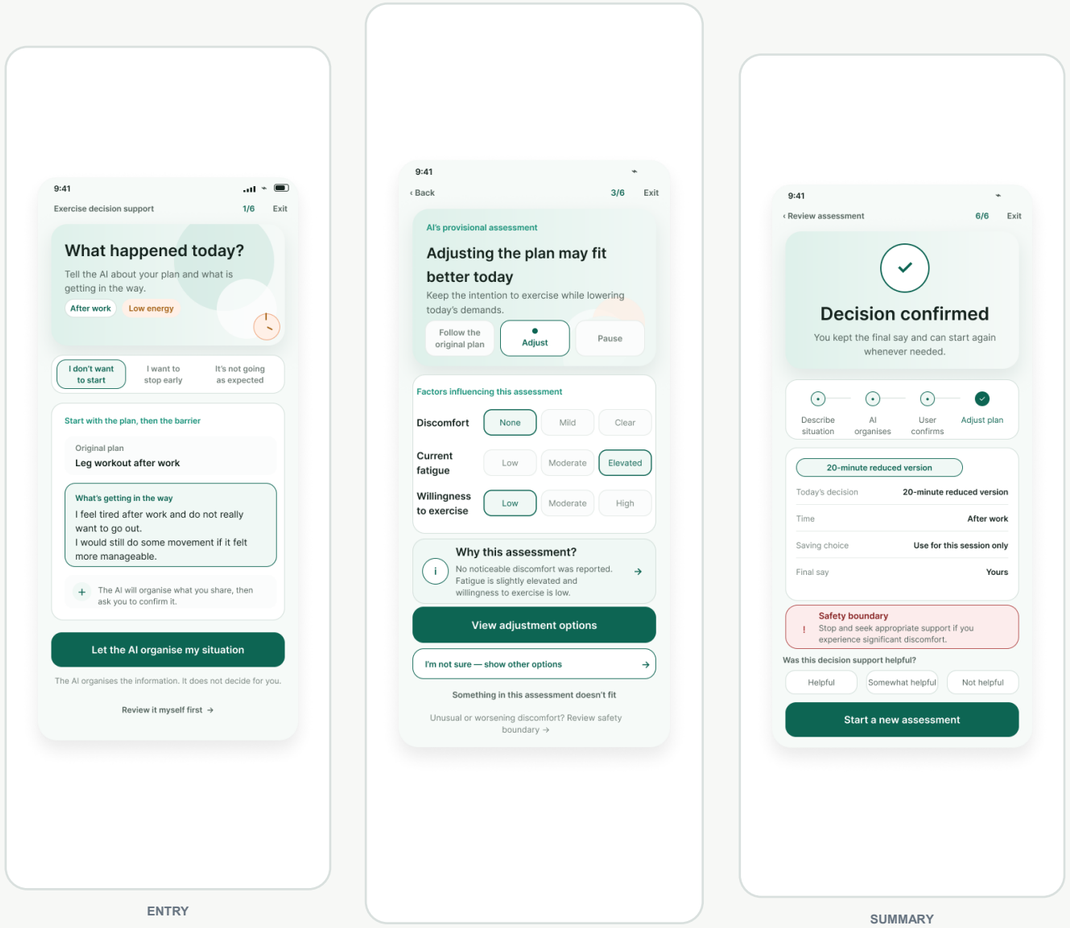
ROLE

Product Designer / UX Designer

FOCUS

AI decision support · willing-to-move moment · user authority

POST-STUDY FIX / AUTHOR QA — NOT PARTICIPANT-VALIDATED



ENTRY JUDGEMENT SUMMARY

The final English screens shown here were frozen after documented author regression QA.

One decision moment, investigated through four forms of evidence.

The project is a high-fidelity preset prototype plus a separate technical feasibility experiment — not a deployed app.

PROJECT PREMISE

When a plan made earlier no longer matches time, energy, motivation or physical state, the task is not adherence. It is deciding what remains appropriate now.

How can AI-assisted decision support help users adapt a planned workout when their current state no longer matches the original plan, while preserving transparency, autonomy and safety?

QUESTIONNAIRE

N=20

26 submissions · 6 excluded

USABILITY

n=10

Round 2 n=4 · P05–P10 n=6

ML HOLDOUT

60

30 EN · 30 ZH synthetic cases

FINAL ENGLISH

AUTHOR QA

Frozen post-study implementation

Evidence types remain separate: questionnaire ≠ usability ≠ ML ≠ author QA.

The plan was made earlier. The decision happens now.

Formative questionnaire evidence shaped the design context; it did not evaluate the final prototype.

08:00 / MORNING PLAN

“Leg day after work.”
The intention is clear, but it is based on a future version of the user.

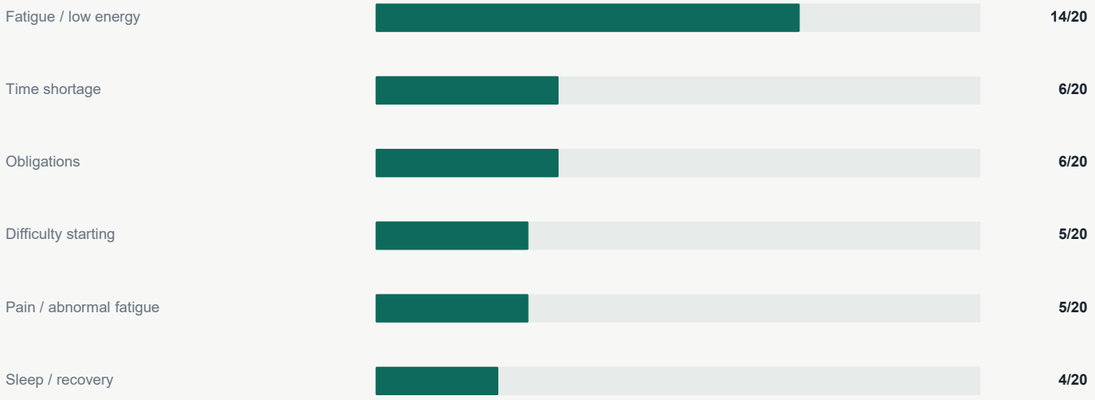
18:30 / CURRENT REALITY

Tired, less willing to go outside, and unsure whether changing the plan means giving up.

THE DECISION GAP

What fits now? Continue, adapt, postpone, decline, or stop because of a safety concern?

N=20 FORMATIVE QUESTIONNAIRE — SELECTED RESPONSE FREQUENCIES



CONTROL REQUESTS

13/20 — correct wrong facts
11/20 — add information + re-evaluate
11/20 — see rationale / uncertainty / alternatives

STOPPING BOUNDARIES

17/20 — discomfort / illness / extreme fatigue
12/20 — explicit refusal
17/20 — refusal should end immediately or after at most one extra step

FORMATIVE · NON-REPRESENTATIVE

Many options were researcher-provided closed choices; frequencies are descriptive selections, not spontaneous themes.

Four research lenses constrain the design.

The literature informed design rationale; it did not supply participant outcomes the project did not collect.

EXERCISE AUTONOMY

Avoid treating adherence as the only success.

Teixeira et al. (2012) · Ntoumanis & Moller (2025)

CONTEXT-SENSITIVE SUPPORT

State changes create decision points, tailoring variables and rules.

Nahum-Shani et al. (2018) · Hardeman et al. (2019)

HUMAN–AI INTERACTION

Interpretation, explanation, correction and control must be designed together.

Amershi et al. (2019) · Miller (2019) · Buçinca et al. (2021)

SAFETY / GOVERNANCE

Risk is contextual and cannot be reduced to one accuracy figure.

WHO (2021) · NIST AI RMF (2023)

LITERATURE → design constraints and interpretation; PARTICIPANTS → observed comprehension and friction.

The project narrowed until one decision became researchable.

Content preserved from the approved previous Case Study.

BROAD EXPLORATION

AI support after exercise setbacks

- missed session
- stopping early
- reduced confidence

CONTEXT-SENSITIVE SUPPORT

What makes the response specific to exercise?

- time
- energy
- motivation
- physical state

ONE DECISION MOMENT

An earlier plan no longer matches the present state

adapt · postpone · decline · safety_stop

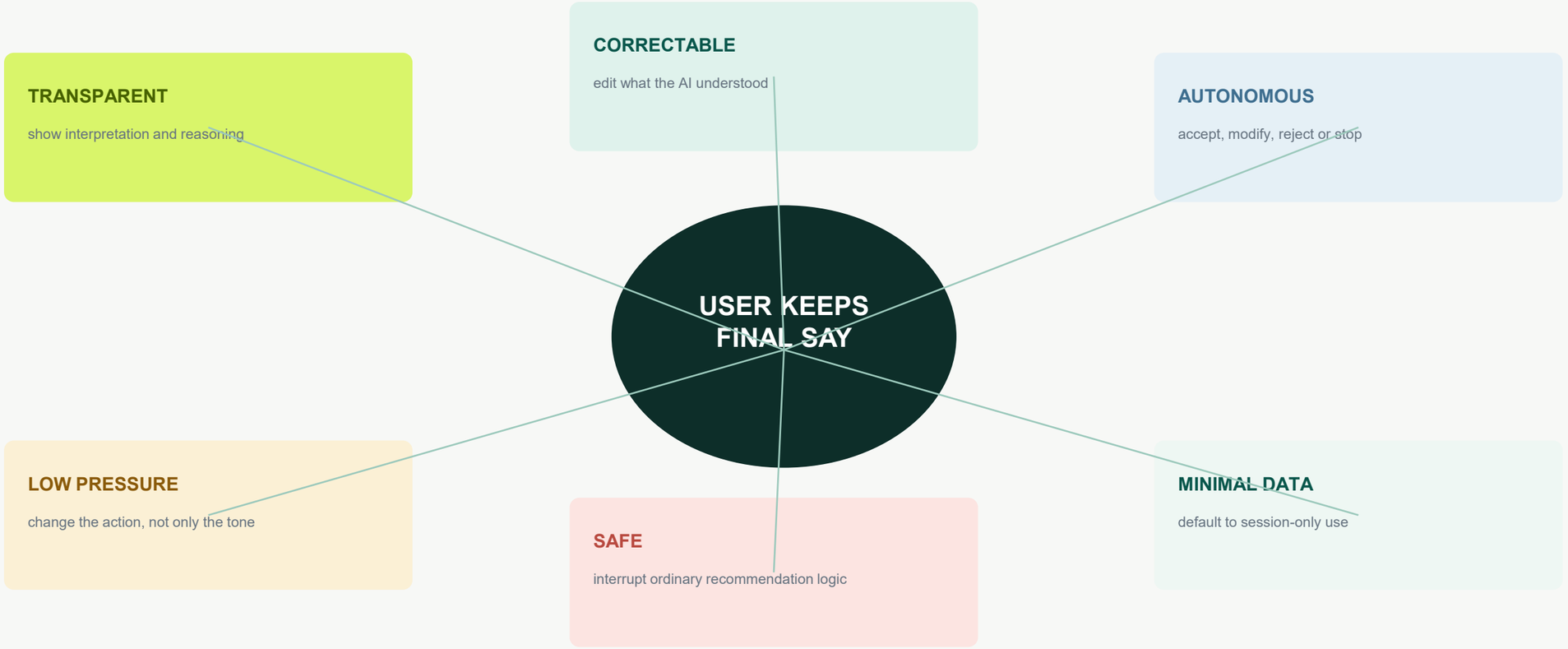
FINAL SCOPE

Support a single uncertain choice without moral ranking.

Not a workout-plan generator. Not performance tracking. Not an adherence-maximisation app.

Six constraints shape one decision.

These are design constraints. They are not evidence that every interaction was validated.



System logic and interface logic answer different questions.

The two evidence streams complement each other but do not validate one another.

HUMAN INTERACTION

What should the user see, question and control?

Interpretation → reasoning → correction → options → autonomy → privacy → usability evidence

Participant testing used Chinese-language content-equivalent prototypes; the frozen English implementation was later author-QA'd.

PARTICIPANT EVIDENCE

TECHNICAL FEASIBILITY

Can the decision route be made inspectable and tested?

Synthetic seed data → MiniLM / TF-IDF retrieval → intent rules → safety gate → frozen evaluation

The ML engine was never connected live to Figma or participant sessions.

FROZEN ML EVIDENCE

NO LIVE INTEGRATION · NO CLAIM THAT CLASSIFIER PERFORMANCE = UX QUALITY

A pragmatic synthetic seed set made the technical route inspectable.

Technical feasibility only: no project-specific training, no fine-tuning, no claim that MiniLM was the “best” model.

REFERENCE SCENARIOS

42

Pragmatic seed-set size; no formal sample-size calculation

CATEGORY SPLIT

11 / 11 / 11 / 9

adapt · postpone · decline · safety_stop

MANUAL REVIEW

42/42

13 texts clarified · zero labels changed

FROZEN CHRONOLOGY

NOT RERUN

Holdout metrics were not recomputed after text clarifications

MINILM — PRAGMATIC SEMANTIC BASELINE

sentence-transformers/all-MiniLM-L6-v2
- pretrained 384-dimensional sentence embeddings
- cosine similarity over normalised vectors
- no training or fine-tuning
- no neural model-selection benchmark

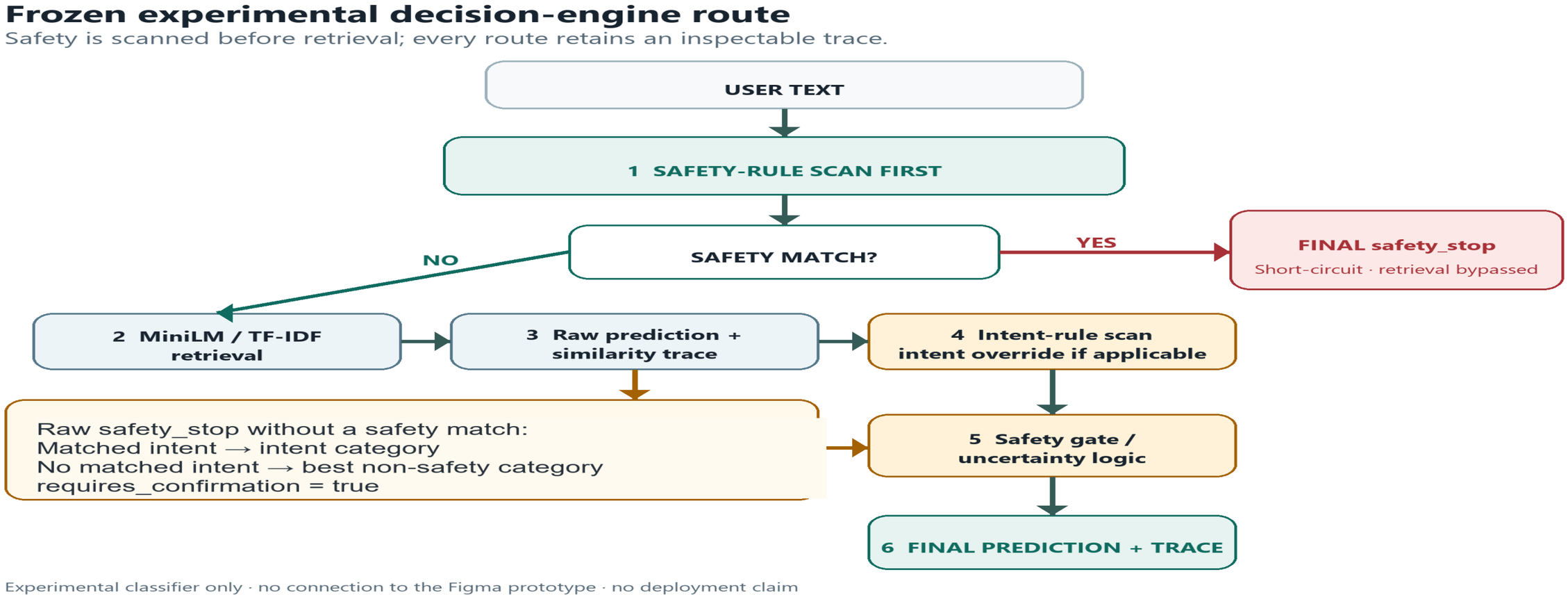
TF-IDF — LEXICAL BASELINE / FALLBACK

Word unigrams + bigrams and character 2–5-grams
- simpler inspectable comparison
- keeps the pipeline runnable if the transformer model cannot load
- tests whether model complexity is actually useful on this small corpus

All scenarios are LLM-assisted synthetic seed data — not participant inputs, not medical ground truth, not representative language.

Safety is scanned before retrieval. Every route retains an inspectable trace.

This architecture is an experimental decision-engine route, separate from the Figma prototype.



EXPLICIT SAFETY MATCH → safety_stop immediately; retrieval is bypassed.

NO SAFETY MATCH → retrieval → intent rules → safety gate / uncertainty → final prediction + trace.

An inspectable rule can change the raw retrieval result.

Hybrid improvement cannot be attributed to the embedding model alone.

RAW RETRIEVAL

Example user meaning:
“I planned leg day after work, but I feel tired and do not want to go outside.”

Nearest-neighbour evidence can mix decline and adapt because the wording carries both reluctance and willingness to reduce demand.

DECLINE

INTENT RULE

Explicit willingness to reduce / shorten / move indoors supports ADAPT.

Rules are transparent interaction boundaries, not medical judgements.

FINAL
ADAPT

Trace keeps both the raw retrieval and the override reason.

The project prioritises inspectability over hiding disagreement between model and rule layers.

The failure changed who can authorise safety_stop.

Safety gating reduced one risk mode but did not make the system clinically safe.

BEFORE SAFETY GATING

Semantic retrieval could produce a final safety_stop even when explicit project-defined safety rules did not fire.

This created false positives such as negated or unrelated “chest” phrases.

AFTER SAFETY GATING

A final safety_stop requires an explicit safety-rule match.

If raw retrieval predicts safety_stop without a safety match, a matched intent determines the non-safety outcome. Otherwise, use the best non-safety retrieval category. In either case, requires_confirmation = true; retain the override trace.

FINAL FALSE POSITIVES

0/45

Non-safety cases in frozen holdout

SAFETY RECALL

14/15

93.33% — one retained false negative

SAFETY PRECISION

1.00

Within the synthetic frozen holdout

RETAINED MISS

H-EN-S07

“pain became suddenly sharp” → adapt

Zero false positives ≠ medical safety. Rule coverage remains incomplete and non-clinical.

Pure retrieval versus safety-gated architecture.

The 70% result is diagnostic and comparative — not a predefined pass score, validation threshold or deployment-readiness threshold.

METHOD	OVERALL	RATE
MiniLM pure	33/60	55.00%
MiniLM gated	42/60	70.00%
TF-IDF pure	38/60	63.33%
TF-IDF gated	42/60	70.00%

LANGUAGE SPLIT

MiniLM pure: EN 22/30 · ZH 11/30
MiniLM gated: EN 22/30 · ZH 20/30
TF-IDF pure: EN 18/30 · ZH 20/30
TF-IDF gated: EN 20/30 · ZH 22/30

INTERPRETATION

Gating improved the project-defined safety boundary, but non-safety distinctions still confused adaptation, postponement and refusal. TF-IDF equalled or exceeded MiniLM in several comparisons.

FROZEN HISTORY: rules / thresholds / holdout inputs / expected labels / recorded outputs were not post-hoc changed to improve this result.

An evidence board, not an internal simulation.

Every technical claim is tied to frozen files, trace fields, tests or retained failures.

WHAT THE ML EVIDENCE SUPPORTS

- an inspectable four-category decision route
- transparent rule / retrieval disagreement
- a documented safety-gating trade-off
- reproducible frozen synthetic evaluation

WHAT IT DOES NOT SUPPORT

- real-world generalisation
- medical correctness
- calibrated probabilities
- behavioural effectiveness
- deployment readiness

RETAINED FAILURE

H-EN-S07 remains wrong in the frozen holdout.

The miss is shown rather than tuned away after observation.

SEPARATION OF CLAIMS

ML metrics describe classification behaviour on synthetic frozen inputs. They do not establish whether users understand, accept or trust the interface.

SYNTHETIC | FROZEN | NON-CLINICAL | NOT DEPLOYED | TRACEABLE

Testing is small, staged and version-bounded.

Pilot, Round 2 and P05–P10 are separate evidence stages; later participants do not retrospectively validate earlier or later versions.

EARLY PILOT

n=1

Reported separately; excluded from formal n=10

ROUND 2

n=4

New participants; early core-flow evidence

SUPPLEMENTARY

n=6

P05–P10; versioned sessions

FORMAL USABILITY

n=10

Round 2 + P05–P10; pilot excluded

VERSION BOUNDARIES

P05–P07

pre-Recovery fix

→

P08

Recovery revised · correction routing still wrong

→

P09

distinct correction editors · post-correction inconsistency

→

P10

consistency revised · sequential state persistence defect

→

Final English

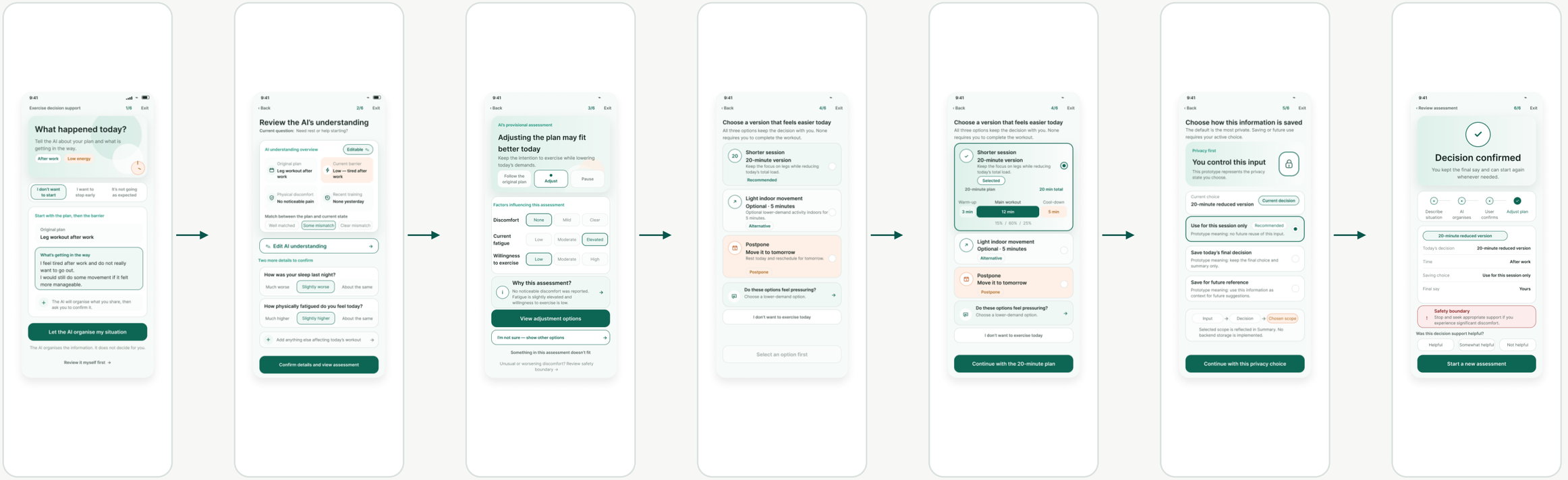
post-study repairs · author regression QA only

Chinese supplementary testing cannot establish the clarity of the final English wording.

Seven screens show the primary preset path.

Frozen Final English walkthrough — shown for assessor context, not as participant-tested English evidence.

POST-STUDY FIX / AUTHOR QA — NOT PARTICIPANT-VALIDATED

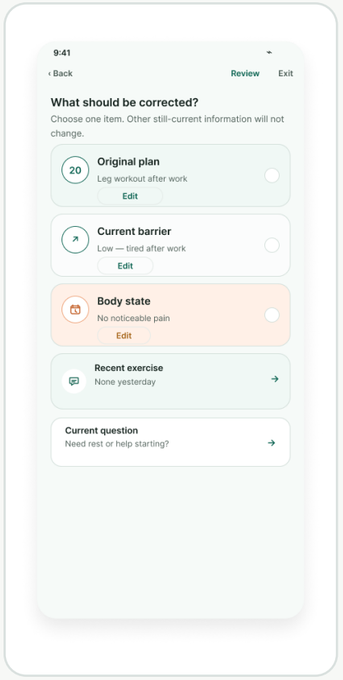


1 · ENTRY | 2 · UNDERSTANDING | 3 · JUDGEMENT | 4 · OPTIONS | 5 · SELECTED | 6 · PRIVACY | 7 · SUMMARY

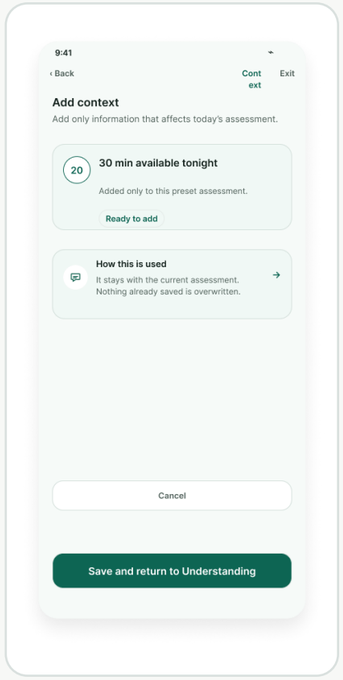
Supplementary main-flow evidence: 5/6 completed independently; P09 completed with moderator assistance.

The user can inspect, challenge and leave the recommendation.

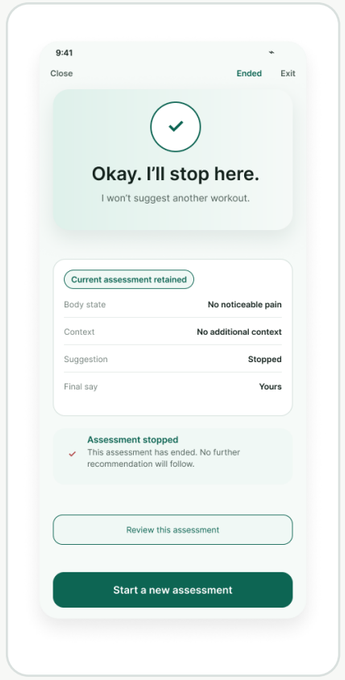
Final English implementation shown below = author QA; participant findings remain tied to the Chinese versions actually tested.



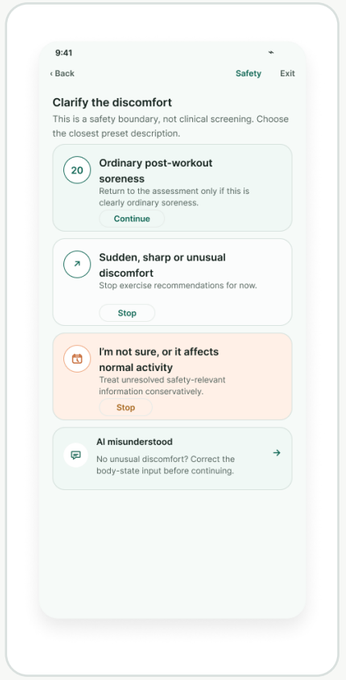
CORRECTION



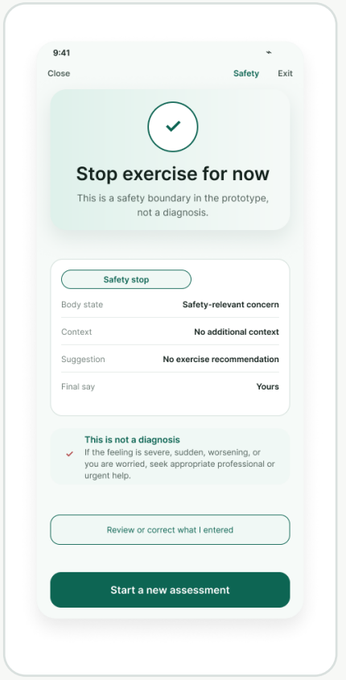
ADD CONTEXT



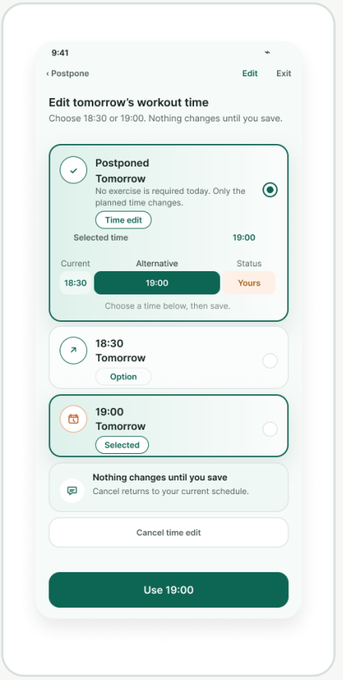
EXPLICIT REFUSAL



SAFETY CLARIFY



SAFETY STOP



TIME EDIT 19:00

VISIBLE INTERPRETATION

CORRECTION / CONTEXT

REFUSAL / ESCAPE

SAFETY BOUNDARY

STATEFUL RE-ENTRY

Author QA covered the frozen English routes. That QA does not become participant validation.

The evidence ledger is small, but specific.

No merged denominators: Time Edit was only tested by P06, P08 and P10; RF3 is one P10 defect, not a group rate.

MAIN FLOW

5/6

independent · P09 assisted

ADVISORY AI

6/6

understood assistive role

FINAL AUTHORITY

6/6

understood user keeps final say

REFUSAL

6/6

understood no continued coercion

PRIVACY

6/6

understood session-only meaning

RF1 · RE-ENTRY

5/6 — P05, P07, P08, P09, P10
Repeated friction when circumstances changed again or users wanted to re-enter after rest / ending.

RF2 · SAFETY SIGNPOSTING

5/6 clean — P06, P07, P08, P09, P10
P05 excluded because moderator explanation influenced the Safety task.

RF3 · STATE PERSISTENCE

1 concrete P10 defect
After a correction plus further context, an edited Body State reverted. Later Final English persistence repair = author QA only.

TIME EDIT

3/3

P06 · P08 · P10 independently

Specific failures produced specific changes.

The project records defects instead of smoothing them into a single “validated” narrative.

P07

Recovery contradiction

Only exercise next steps remained after a recovery judgement.

→

P08

Correction routing

All five Modify controls led to one generic editor.

→

P09

Correction consistency

Distinct editors worked, but return states were visually / interactively inconsistent.

→

P10

Sequential persistence

After correction + added context, edited Body State reverted.

→

POST-STUDY

Final English repairs

Recovery / editors / persistence / refusal / pressure / safety / privacy / reset author-QA’d.

VERIFIED P07 QUOTATION — ENGLISH TRANSLATION FROM CHINESE

“I had already indicated that my condition was much worse, and the advice was to rest first and consider exercise later. But on the next page there were only three options, all about exercise. This created a contradiction: I clearly needed to rest first, but the next step still only allowed me to choose exercise. Regardless of intensity, it was still exercise, which is different from rest.”

Only preserved verbatim participant quotations are used as quotations; unrecorded comments remain researcher paraphrases.

Participant-tested evidence and Final English author QA are different categories.

Historical Chinese testing is not rewritten to match the later English implementation.

What was tested, and what was fixed later

The two evidence states are related but must not be merged.

A · PARTICIPANT-TESTED EVIDENCE



Chinese-language, content-equivalent prototype

Round 2
n = 4

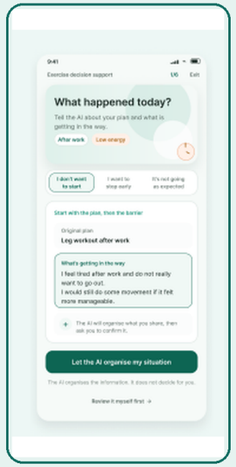
Supplementary P05–P10
n = 6

Formal usability total
n = 10 (pilot excluded)

Pilot n = 1, reported separately
Evidence remains version-specific.

Image: tested Chinese interface state
(not a participant photograph)

B · FINAL ENGLISH PROTOTYPE



Figma page
90_FINAL · EN SUBMISSION

Entry
V15_EN_01_Entry

Canonical node
687:5410

Frozen after documented
author regression QA

POST-STUDY FIX / AUTHOR QA
NOT PARTICIPANT-VALIDATED

A · PARTICIPANT-TESTED CHINESE

Round 2 n=4 + supplementary P05–P10 n=6. Evidence applies to the version each participant encountered.

B · VERSION-SPECIFIC DEFECTS

RF1 / RF2 / RF3 and the P07 quotation document friction and failures across changing prototype states.

C · FROZEN FINAL ENGLISH

POST-STUDY FIX / AUTHOR QA. Regression-tested by the author; NOT PARTICIPANT-VALIDATED.

The ML experiment remains a fourth, separate evidence type.

What the evidence supports — and what it does not.

The strongest submission claim is bounded design and technical feasibility, not product effectiveness.

SUPPORTED WITHIN THIS STUDY

01 · Formative questionnaire context from valid N=20

02 · Supplementary main-flow completion: 5/6 independent, P09 assisted

03 · 6/6 comprehension of advisory role, final authority, refusal and session-only privacy

04 · Time Edit 3/3 among P06 / P08 / P10

05 · RF1 = 5/6 · RF2 = 5/6 clean · RF3 = one P10 defect

06 · Frozen synthetic ML holdout and retained H-EN-S07 miss

NOT ESTABLISHED

- behavioural effectiveness
- reduced psychological pressure as a measured effect
- clinical safety or medical correctness
- real personalised recommendation quality
- longitudinal reliability / adherence effects
- deployment readiness
- participant validation of the frozen Final English repairs

Contribution: a transparent, revisable decision-support interaction + an inspectable technical feasibility route + a documented failure history.

Move from reviewed boundaries to controlled validation.

Future work should test the frozen repairs directly rather than inferring success from author QA.

01 · TEST FROZEN ENGLISH

Run participant regression on the frozen Final English correction, refusal, pressure, safety, privacy, reset and persistence routes.

02 · RF3 PERSISTENCE

Repeat sequential correction + context changes to verify state persistence under realistic re-entry sequences.

03 · RF1 RE-ENTRY

Evaluate return-to-assessment after rest, changed mind or new information without forcing users to restart.

04 · RF2 SIGNPOSTING

Test non-diagnostic next-step language after Safety Stop without drifting into diagnosis or prescribing.

05 · ACCESSIBILITY + LONGITUDINAL

Run systematic accessibility checks and longer-term use studies before any effectiveness or reliance claim.

06 · TECHNICAL ROBUSTNESS

Use representative real-user language, preserve H-EN-S07 as a regression case, and evaluate safety rules beyond the small synthetic corpus.

NO DEPLOYMENT CLAIM UNTIL interaction evidence, data quality, safety handling, accessibility and real-world reliability are substantially stronger.